IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART IV: MEng and ACGI

WAVELETS, REPRESENTATION LEARNING AND THEIR APPLICATIONS

Thursday, 16 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : A. Manikas

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WAVELETS AND APPLICATIONS

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Special Information for the Invigilators: NONE

Information for Candidates:

Sub-sampling by an integer N

$$x_{\downarrow N}[n] \Longleftrightarrow \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j(\omega-2\pi k)/N}) = \frac{1}{N}\sum_{k=0}^{N-1} X(W_N^k z^{1/N}),$$

where

$$W_N^k = e^{-j2\pi k/N}.$$

The operators "subsampling by N" ($\downarrow N$) and "upsampling by M" ($\uparrow M$) commute if and only if N and M are relatively prime. For example subsampling by 2 followed by upsampling by 3 commute since two and three are co-prime numbers.

Dual Basis:
Given a basis $\{\varphi_i(t)\}_{i\in\mathbb{Z}}$, the dual basis is given by the set of elements $\{\tilde{\varphi}_i(t)\}_{i\in\mathbb{Z}}$ satisfying:

$$\langle \varphi_i(t), \tilde{\varphi}_j(t)\rangle = \delta_{i,j}.$$

The Questions

1. Consider the two-channel filter bank of Figure 1a.

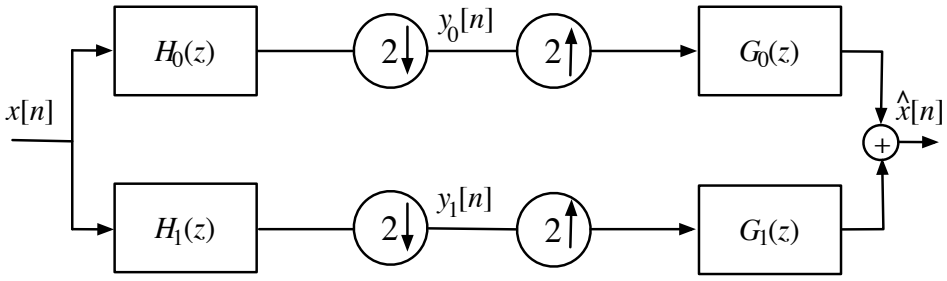


Figure 1a: Two-channel filter bank.

(a) Express $\hat{X}(z)$ as a function of $X(z)$ and the filters. Then derive the two perfect reconstruction (PR) conditions the filters have to satisfy. [5]

(b) Assume that $G_0(z) = H_1(-z)$ and $G_1(z) = -H_0(-z)$.

i. Show that, under these conditions, the aliased component, $X(-z)$, of $x[n]$ is removed. [4]

ii. Further assume $H_0(z) = (1 + z^{-1})^2$ and $H_1(z) = -(1 - z^{-1})^2(1 + 4z^{-1} + z^{-2})$, find $\hat{x}[n]$ for the following inputs

A. $x[n] = \delta[n]$, [4]

B. $x[n] = (-1)^n$. [4]

(c) Assume now that $H_0(z) = (1 + z)/\sqrt{2}$ and $G_0(z) = (1 + z^{-1})/\sqrt{2}$.

i. Show that $P(z) = H_0(z)G_0(z)$ satisfies $P(z) + P(-z) = 2$. [4]

ii. Design the filters $H_1(z)$ and $G_1(z)$ in order to have a perfect reconstruction orthogonal filter bank. [4]

2. (a) Consider the system shown in Figure 2a with input $x[n]$ and output $y[n]$.

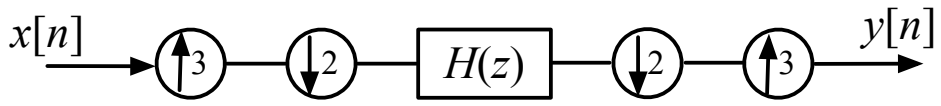


Figure 2a: Multi-rate system.

Assume that an equivalent representation of the above system is given in Fig 2b.

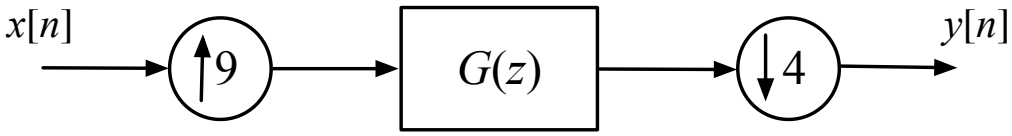


Figure 2b: A multi-rate system equivalent to the one in Fig. 2a

Given that $H(z) = z^{-2}$, find the exact expression for $G(z)$. [5]

(b) Consider the system

$$y = HD_3GHD_2HFD_2x$$

where G, H and F are the infinite matrix representations for filtering and D_k is the matrix representation of downsampling by k with $k = 2, 3$.

i. Draw a block diagram of the system. [5]

ii. Derive an equivalent system consisting of a single filter block and a single downsampling matrix. Write the z-transform of the equivalent system as a function of $G(z)$, $H(z)$ and $F(z)$. [5]

Question continues on next page.

(c) With reference to Fig. 2c, given an input $x[n]$, consider upsampling by 2 followed by interpolation with a filter $H(z)$. Then to recover the original signal, apply filtering with a filter $G(z)$ followed by downsampling by 2 in order to obtain a reconstruction $\hat{x}[n]$.

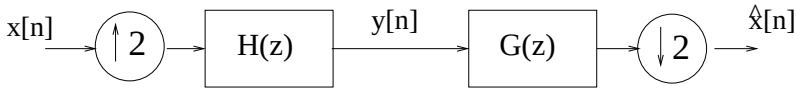


Figure 2c: Interpolation followed by decimation.

i. What does the product $P(z) = H(z)G(z)$ have to satisfy in order for $\hat{x}[n]$ to be equal to $x[n]$? [5]

ii. Assume that

$$H(z) = \frac{(1 + z^{-1})(1 + z)}{4}.$$

Find a 5-tap symmetric filter $G(z)$ such that perfect reconstruction is achieved. Please note that the solution is not unique. [5]

3. Consider the interval $t \in [0,3]$ and let

$$\varphi_1(t) = \begin{cases} t, & \text{for } t \in [0,1] \\ 2-t & \text{for } t \in (1,2] \\ 0, & \text{for } t \in (2,3]. \end{cases}$$

Denote with $V = span(\{\varphi_1(t), \varphi_2(t), \varphi_3(t)\})$ the sub-space generated by $\varphi_1(t)$ and its circular shifts by 1 over the interval $t \in [0,3]$. The three basis functions are shown in Fig. 3.

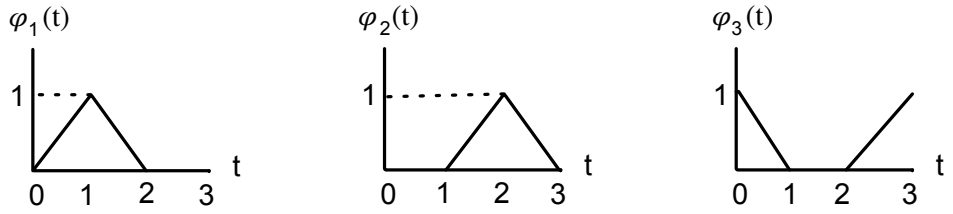


Figure 3: Three functions $\varphi_1, \varphi_2, \varphi_3$ defined for $t \in [0,3]$ and related by circular shifts.

Given a signal $x(t)$ defined for $t \in [0,3]$, the aim is to compute the orthogonal projection of $x(t)$ onto V. Recall that this is given by:

$$x_v(t) = \sum_{i=1}^{3} \langle x(t), \tilde{\varphi}_i(t) \rangle \varphi_i(t)$$

where $\{\tilde{\varphi}_i(t)\}_{i=1}^3$ are the three dual-basis functions.

(a) Since $\tilde{\varphi}_i(t) \in V$ we can write $\tilde{\varphi}_i(t) = \sum_{k=1}^3 \alpha_{i,k} \varphi_k(t)$. Using this fact

i. Determine the three dual-basis functions $\tilde{\varphi}_i(t)$, $i = 1,2,3$. That is, find the coefficients $\alpha_{i,k}$, $i = 1,2,3$; $k = 1,2,3$. [8]

ii. Sketch and dimension $\tilde{\varphi}_i(t)$ $i = 1,2,3$. [7]

(b) Given the dual basis and the signal

$$x(t) = \begin{cases} 1, & \text{for } t \in [0,2] \\ 0 & \text{for } t \in (2,3]. \end{cases}$$

i. Compute the inner products $\langle x(t), \tilde{\varphi}_i(t) \rangle$, $i = 1,2,3$. [5]

ii. Verify that the error $\epsilon(t) = x(t) - x_v(t)$ is orthogonal to V. [5]

4. (a) Consider an arbitrary real-valued function $\varphi(t)$ with Fourier transform $\hat{\varphi}(\omega)$. Show that the Fourier transform of the sequence $a[n] = \langle \varphi(t), \varphi(t-n) \rangle$ is:

$$A(e^{j\omega}) = \sum_{k=-\infty}^{\infty} |\hat{\varphi}(\omega + 2\pi k)|^2.$$

[5]

(b) Consider the linear B-Spline given by

$$\varphi(t) = \begin{cases} t, & 0 \leq t < 1 \\ 2 - t, & 1 \leq t < 2 \\ 0, & \text{otherwise.} \end{cases}$$

We know that $\varphi(t)$ is a valid scaling function. Therefore, it satisfies the Riesz basis criterion:

$$0 < A \leq \sum_{n \in \mathbb{Z}} |\hat{\varphi}(\omega + 2\pi n)|^2 \leq B < +\infty.$$

Compute the bounds A and B.

[5]

(c) Suppose you are given a two-channel FIR filter bank with real coefficients and synthesis lowpass filter

$$g_0[n] = \frac{1}{4\sqrt{2}}(\delta_n + 3\delta_{n-1} + 3\delta_{n-2} + \delta_{n-3}).$$

Consider the equivalent filter

$$G_0^{(i)}(z) = \prod_{k=0}^{i-1} G_0(z^{2^k})$$

obtained by iterating the filter bank decomposition i times. Consider the function

$$\varphi^{(i)}(t) = 2^{i/2} g_0^{(i)}[n], \qquad n/2^i \leq t < (n+1)/2^i.$$

i. Can you say anything about the convergence of $\lim_{i \to \infty} \varphi^{(i)}(t)$?

[5]

ii. We know that, in the case of convergence, $\varphi(t)$ is a valid scaling function. Can you say anything about continuity of this function?

[5]

iii. State the number of vanishing moments of the analysis wavelet function obtained from $\varphi(t)$.

[5]